

理论力学作业

1900011604 张植竣 北京大学

2020 年 10 月 23 日

第一章

1.(a) A 点坐标为:

$$(l_1 \cos \theta, l_1 \sin \theta)$$

由勾股定理, B 点坐标表示为:

$$(l_1 \cos \theta + \sqrt{l_2^2 - l_1^2 \sin^2 \theta}, 0)$$

1.(b) 由虚功原理得:

$$P \cdot l_1 \cos \theta d\theta - Q \left(l_1 \sin \theta d\theta + \frac{l_1^2 \sin \theta \cos \theta}{\sqrt{l_2^2 - l_1^2 \sin^2 \theta}} \right) = 0$$

故解得:

$$\frac{Q}{P} = \tan \theta + \frac{l_1 \sin \theta}{\sqrt{l_2^2 - l_1^2 \sin^2 \theta}}$$

2. 弹性绳伸长量原来为:

$$\Delta x = 2\pi R \sin \theta - l_0$$

以球心为原点, 竖直向上为正方向, 则弹性绳高度为:

$$y = R \cos \theta$$

由虚功原理得:

$$k(2\pi R \sin \theta - l_0) \cdot 2\pi R \cos \theta d\theta + \sigma l_0 g \cdot (-R \sin \theta d\theta)$$

解得 θ 是下列超越方程的解:

$$\sigma l_0 g \tan \theta = 2\pi k(2\pi R \sin \theta - l_0), \quad \theta \in (0, \frac{\pi}{2})$$

第二章

1. 由运动学方程 $v^2 - v_0^2 = 2ax$ 及 $ds = \sqrt{1 + (y')^2}dx$ 得:

$$dt = \frac{\sqrt{1 + (y')^2}dx}{\sqrt{2gy}}$$

得到总时间 T 作为运动曲线的泛函:

$$\begin{aligned} T &= \int_{x_1}^{x_2} F(y, y', x) dx \\ &= \int_{x_1}^{x_2} \frac{\sqrt{1 + (y')^2}}{\sqrt{2gy}} dx \end{aligned}$$

则最速降线要求:

$$\delta F = 0$$

可以发现: $F(y, y', x)$ 不显含 x , 即:

$$\frac{\partial F}{\partial x} = 0$$

考虑到时间平移不变性蕴含着能量守恒, 对于泛函 $F(y, y', x)$, 应该有一个类似的守恒量:

$$E = y' \frac{\partial F}{\partial y'} - F$$

保持为常数 C , 即:

$$\begin{aligned} y' \frac{\partial F}{\partial y'} - F &= y' \cdot \frac{y'}{\sqrt{2gy[1 + (y')^2]}} - \frac{\sqrt{1 + (y')^2}}{\sqrt{2gy}} \\ &= C \end{aligned}$$

分离变量得:

$$\sqrt{\frac{2gC^2y}{1 - 2gC^2y}} dy = dx$$

令:

$$k = \frac{1}{4gC^2}$$

化简得到该曲线满足的微分方程:

$$\sqrt{\frac{y}{2k - y}} dy = dx$$

考虑到:

$$\begin{aligned} \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} &= \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} \sqrt{\frac{1 - \cos \theta}{1 - \cos \theta}} \\ &= \sqrt{\frac{(1 - \cos \theta)^2}{1 - \cos^2 \theta}} \\ &= \frac{1 - \cos \theta}{\sin \theta} \end{aligned}$$

令:

$$\frac{y}{2k - y} = \frac{1 - \cos \theta}{1 + \cos \theta}$$

则可以解出：

$$y = k(1 - \cos \theta)$$

$$\begin{aligned} \sqrt{\frac{y}{2k-y}} dy &= \sqrt{\frac{k(1-\cos\theta)}{k(1+\cos\theta)}} k \sin \theta d\theta \\ &= \frac{1-\cos\theta}{\sin\theta} k \sin \theta d\theta \\ &= k(1-\cos\theta) d\theta \end{aligned}$$

即：

$$dx = k(1 - \cos \theta) d\theta$$

积分得：

$$x = k(\theta - \sin \theta) + C_1$$

由 $x = 0$ 时 $\theta = 0$ ，定出 $C_1 = 0$ ，则解出：

$$\begin{cases} x = k(\theta - \sin \theta) \\ y = k(1 - \cos \theta) \end{cases}$$

这是一条摆线。

2. 由体系总能量（重力势能）最小得：：

$$V = \int_{x_1}^{x_2} \lambda \sqrt{1 + (y')^2} g y \, dx$$

则最速降线要求：

$$\delta V = 0$$

可以发现： $V(y, y', x)$ 不显含 x ，即：

$$\frac{\partial V}{\partial x} = 0$$

考虑到时间平移不变性蕴含着能量守恒，对于泛函 $E(y, y', x)$ ，应该有一个类似的守恒量：

$$E = y' \frac{\partial V}{\partial y'} - V$$

保持为常数 C ，即：

$$\begin{aligned} y' \frac{\partial V}{\partial y'} - V &= y' \cdot \frac{\lambda g y y'}{\sqrt{1 + (y')^2}} - \lambda \sqrt{1 + (y')^2} g y \\ &= C \end{aligned}$$

分离变量得：

$$\sqrt{\frac{C^2}{\lambda^2 g^2 y^2 - C^2}} dy = dx$$

令：

$$k = \frac{\lambda g}{C}$$

化简得到该曲线满足的微分方程：

$$\frac{1}{\sqrt{k^2 y^2 - 1}} dy = dx$$

两边积分得：

$$\operatorname{arccosh} ky = kx + C_1$$

即：

$$y = \frac{1}{k} \cosh(kx + C_1)$$

由两端悬挂点等高，即 $y(x) = y(-x)$ ，定出 $C_1 = 0$ ，则有：

$$y = \frac{1}{k} \cosh kx$$

由绳长为 L ，两端悬挂点之间距离为 b ，得：

$$\begin{aligned} \int_{-\frac{b}{2}}^{\frac{b}{2}} \sqrt{1 + (y')^2} dx &= \int_{-\frac{b}{2}}^{\frac{b}{2}} \sqrt{1 + \sinh^2 kx} dx \\ &= \int_{-\frac{b}{2}}^{\frac{b}{2}} \cosh kx dx \\ &= 2 \sinh \frac{kb}{2} \\ &= L \end{aligned}$$

解得：

$$k = \frac{\lambda g}{C} = \frac{2}{b} \operatorname{arcsinh} \frac{L}{2}$$

综上，绳子平衡时构成的曲线的形状为：

$$y = \frac{b}{2} \cosh \frac{2}{b} \frac{\operatorname{arcsinh} \frac{L}{2} x}{\operatorname{arcsinh} \frac{L}{2}}$$

3.(a) 三维形式下自由粒子的作用量为：

$$\begin{aligned} S &= -mc \int ds \\ &= -mc \int \sqrt{c^2 dt^2 - d\mathbf{x}^2} \\ &= -mc^2 \int \sqrt{1 - \frac{\mathbf{v}^2}{c^2}} dt \end{aligned}$$

则 Lagrange 量：

$$L(\mathbf{v}) = -mc^2 \sqrt{1 - \frac{\mathbf{v}^2}{c^2}}$$

由 Euler-Lagrange 方程得到：

$$\frac{\partial L}{\partial \mathbf{x}} = \frac{d}{dt} \left(\frac{\partial L}{\partial \mathbf{v}} \right) = 0$$

而粒子的广义动量为:

$$\mathbf{p} = \frac{\partial L}{\partial \mathbf{v}} = \frac{m\mathbf{v}}{\sqrt{1 - \frac{v^2}{c^2}}}$$

则粒子的运动方程为:

$$\frac{d\mathbf{p}}{dt} = 0$$

即粒子做匀速直线运动。

3.(b) 四维形式下自由粒子的作用量为:

$$\begin{aligned} S &= -mc \int ds \\ &= -mc \int \sqrt{dx_\mu dx^\mu} \end{aligned}$$

利用变分法, 有:

$$\begin{aligned} \delta S &= -mc \int \delta \sqrt{dx_\mu dx^\mu} \\ &= -mc \int \frac{\delta(dx_\mu)dx^\mu + \delta(dx^\mu)dx_\mu}{2\sqrt{dx_\mu dx^\mu}} \\ &= -mc \int \frac{\delta(dx_\mu)dx^\mu}{\sqrt{dx_\mu dx^\mu}} \\ &= -mc \int \frac{dx^\mu}{ds} d(\delta x_\mu) \\ &= 0 \end{aligned}$$

分部积分得:

$$-mc \int \frac{dx^\mu}{ds} d(\delta x_\mu) = -mc \left. \frac{dx^\mu}{ds} \delta x_\mu \right|_{\tau_2}^{\tau_1} + mc \int \frac{d^2 x^\mu}{ds^2} \delta x_\mu ds = 0$$

第一项是边界项, 在端点处要求 $\delta x_\mu = 0$, 则边界项为零。

第二项中, 由于 δx_μ 是任意的变分, 粒子的运动方程为:

$$\frac{d^2 x^\mu}{ds^2} = 0$$

即粒子做匀速直线运动。

5.(a) 设标量外场为 $\Phi(x)$, 则一个相对论性粒子的作用量为:

$$S = -mc \int e^{\Phi(x)} ds$$

利用变分法, 有:

$$\begin{aligned} \delta S &= -mc \int e^{\Phi(x)} ds \\ &= -mc \int [(\delta ds)e^\Phi + (d e^\Phi)ds] \\ &= -mc \int \left[e^\Phi \frac{dx_\mu}{ds} d(\delta x^\mu) + e^\Phi \frac{\partial \Phi}{\partial x^\mu} \delta x^\mu ds \right] \\ &= 0 \end{aligned}$$

分部积分得：

$$\begin{aligned}
& -mc \int \left[e^\Phi \frac{dx_\mu}{ds} d(\delta x^\mu) + e^\Phi \frac{\partial \Phi}{\partial x^\mu} \delta x^\mu ds \right] \\
& = e^\Phi \frac{dx^\mu}{ds} \Big|_{\tau_2}^{\tau_1} + mc \int \left[\delta x^\mu d \left(e^\Phi \frac{dx_\mu}{ds} \right) - e^\Phi \frac{\partial \Phi}{\partial x^\mu} \delta x^\mu ds \right] \\
& = e^\Phi \frac{dx^\mu}{ds} \Big|_{\tau_2}^{\tau_1} + mc \int \left[\delta x^\mu \frac{dx_\mu}{ds} d(e^\Phi) + \delta x^\mu e^\Phi d \left(\frac{dx_\mu}{ds} \right) - e^\Phi \frac{\partial \Phi}{\partial x^\mu} \delta x^\mu ds \right]
\end{aligned}$$

第一项是边界项，在端点处要求 $\delta x^\mu = 0$ ，则边界项为零。

第二项中，代入：

$$\begin{aligned}
d(e^\Phi) & = e^\Phi d\Phi \\
& = e^\Phi \frac{\partial \Phi}{\partial x^\nu} dx^\nu \\
& = e^\Phi \frac{\partial \Phi}{\partial x^\nu} \frac{dx^\nu}{ds} ds
\end{aligned}$$

以及：

$$\begin{aligned}
d \left(\frac{dx_\mu}{ds} \right) & = \frac{d}{ds} \left(\frac{dx_\mu}{ds} \right) ds \\
& = \frac{d^2 x_\mu}{ds^2} ds
\end{aligned}$$

化简得：

$$\begin{aligned}
& mc \int \left[\delta x^\mu \frac{dx_\mu}{ds} d(e^\Phi) + \delta x^\mu e^\Phi d \left(\frac{dx_\mu}{ds} \right) - e^\Phi \frac{\partial \Phi}{\partial x^\mu} \delta x^\mu ds \right] \\
& = mc \int \left[\delta x^\mu \frac{dx_\mu}{ds} e^\Phi \frac{\partial \Phi}{\partial x^\nu} \frac{dx^\nu}{ds} ds + \delta x^\mu e^\Phi \frac{d^2 x_\mu}{ds^2} ds - e^\Phi \frac{\partial \Phi}{\partial x^\mu} \delta x^\mu ds \right] \\
& = mc \int \left[\frac{\partial \Phi}{\partial x^\nu} \frac{dx^\nu}{ds} \frac{dx_\mu}{ds} ds + \frac{d^2 x_\mu}{ds^2} - \frac{\partial \Phi}{\partial x^\mu} \right] \delta x^\mu e^\Phi ds \\
& = 0
\end{aligned}$$

由于 $e^\Phi > 0$ ，且 δx^μ 是任意的变分，得到粒子的运动方程：

$$\frac{\partial \Phi}{\partial x^\nu} \frac{dx^\nu}{ds} \frac{dx_\mu}{ds} ds + \frac{d^2 x_\mu}{ds^2} = \frac{\partial \Phi}{\partial x^\mu}$$

5.(b) 设标量外场为 $\Phi(x)$ ，则一个相对论性粒子的 Lagrange 量为：

$$L = -mc^2 \sqrt{1 - \frac{\mathbf{v}^2}{c^2}} e^{\Phi(\mathbf{x}, t)}$$

由 Euler-Lagrange 方程得到：

$$\begin{aligned}
\frac{\partial L}{\partial \mathbf{x}} & = \frac{d}{dt} \left(\frac{\partial L}{\partial \mathbf{v}} \right) \\
-mc^2 \sqrt{1 - \frac{\mathbf{v}^2}{c^2}} e^\Phi \cdot \frac{\partial \Phi}{\partial \mathbf{x}} & = \frac{d}{dt} \left(-\frac{m\mathbf{v}}{\sqrt{1 - \frac{\mathbf{v}^2}{c^2}}} e^\Phi \right)
\end{aligned}$$

代入粒子的广义动量:

$$\mathbf{p} = \frac{\partial L}{\partial \mathbf{v}} = \frac{m\mathbf{v}}{\sqrt{1 - \frac{v^2}{c^2}}}$$

化简为:

$$mc^2 \sqrt{1 - \frac{v^2}{c^2}} \mathbf{e}^\Phi \cdot \frac{\partial \Phi}{\partial \mathbf{x}} = \mathbf{p} \cdot \mathbf{e}^\Phi \frac{\partial \Phi}{\partial t} + \mathbf{e}^\Phi \frac{d\mathbf{p}}{dt}$$

6. 设矢量外场 $A_\mu(x)$, 则一个相对论性粒子的作用量为:

$$S = -mc \int ds - \frac{e}{c} \int A_\mu(x) dx^\mu$$

利用变分法, 有:

$$\begin{aligned} \delta S &= -mc \int \delta \sqrt{dx_\mu dx^\mu} - \frac{e}{c} \int \delta(A_\mu(x) dx^\mu) \\ &= -mc \int \frac{\delta(dx_\mu) dx^\mu + \delta(dx^\mu) dx_\mu}{2\sqrt{dx_\mu dx^\mu}} - \frac{e}{c} \int [\delta A_\mu dx^\mu + A_\mu d(\delta x^\mu)] \\ &= -mc \int \frac{\delta(dx^\mu) dx_\mu}{\sqrt{dx_\mu dx^\mu}} - \frac{e}{c} \int [\delta A_\nu dx^\nu + A_\mu d(\delta x^\mu)] \\ &= -mc \int \frac{dx_\mu}{ds} d(\delta x^\mu) - \frac{e}{c} \int \delta A_\nu dx^\nu - \frac{e}{c} \int A_\mu d(\delta x^\mu) \\ &= 0 \end{aligned}$$

将含有 $d(\delta x^\mu)$ 的两项分部积分:

$$\begin{aligned} -mc \int \frac{dx_\mu}{ds} d(\delta x^\mu) &= -mc \left. \frac{dx_\mu}{ds} \delta x^\mu \right|_{\tau_2}^{\tau_1} + mc \int \frac{d^2 x_\mu}{ds^2} \delta x^\mu ds = 0 \\ -\frac{e}{c} \int A_\mu d(\delta x^\mu) &= \frac{e}{c} \left. A_\mu \delta x^\mu \right|_{\tau_2}^{\tau_1} + \frac{e}{c} \int \delta x^\mu dA_\mu = 0 \end{aligned}$$

它们的第一项是边界项, 在端点处要求 $\delta x_\mu = 0$, 则边界项为零。

代入 δS 得:

$$\begin{aligned} \delta S &= mc \int \frac{d^2 x_\mu}{ds^2} \delta x^\mu ds - \frac{e}{c} \int \delta A_\nu dx^\nu + \frac{e}{c} \int \delta x^\mu dA_\mu \\ &= mc \int \frac{d^2 x_\mu}{ds^2} \delta x^\mu ds - \frac{e}{c} \int (\partial_\mu A_\nu \delta x^\mu) dx^\nu + \frac{e}{c} \int \delta x^\mu (\partial_\nu A_\mu dx^\nu) \\ &= mc \int \frac{d^2 x_\mu}{ds^2} \delta x^\mu ds - \frac{e}{c} \int \partial_\mu A_\nu \frac{dx^\nu}{ds} \delta x^\mu ds + \frac{e}{c} \int \partial_\nu A_\mu \frac{dx^\nu}{ds} \delta x^\mu ds \\ &= mc \int \frac{d^2 x_\mu}{ds^2} \delta x^\mu ds - \frac{e}{c} \int (\partial_\mu A_\nu - \partial_\nu A_\mu) \frac{dx^\nu}{ds} \delta x^\mu ds \\ &= 0 \end{aligned}$$

由于 δx^μ 是任意的变分, 得到粒子的运动方程:

$$mc \frac{d^2 x_\mu}{ds^2} = \frac{e}{c} (\partial_\mu A_\nu - \partial_\nu A_\mu) \frac{dx^\nu}{ds}$$

记四维张量场强:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

则运动方程写为:

$$mc \frac{d^2 x_\mu}{ds^2} = \frac{e}{c} F_{\mu\nu} \frac{dx^\nu}{ds}$$

9.(a) 系统的动能:

$$T = \frac{1}{2} m \Omega^2 R^2 \sin^2 \theta + \frac{1}{2} m R^2 \dot{\theta}^2$$

系统的势能:

$$V = mgR \cos \theta$$

则系统的 Lagrange 量为:

$$L = T - V = \frac{1}{2} m \Omega^2 R^2 \sin^2 \theta + \frac{1}{2} m R^2 \dot{\theta}^2 - mgR \cos \theta$$

9.(b) 由 Euler-Lagrange 方程得 θ 满足的运动方程为:

$$\frac{\partial L}{\partial \theta} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right)$$

$$m \Omega^2 R^2 \sin \theta \cos \theta + mgR \sin \theta = m R^2 \frac{d\dot{\theta}}{dt}$$

化简为:

$$\Omega^2 R \sin \theta \cos \theta + g \sin \theta = R \ddot{\theta}$$

若使 θ 平衡, 即 $\ddot{\theta} = 0$, 分以下两种情况:

$$\sin \theta = 0$$

$$\cos \theta = -\frac{g}{\Omega^2 R}$$

则平衡位置为:

$$\begin{cases} \theta = 0, \theta = \pi, & \Omega < \sqrt{\frac{g}{R}} \\ \theta = 0, \theta = \pi, \theta = \arccos\left(-\frac{g}{\Omega^2 R}\right), & \Omega \geq \sqrt{\frac{g}{R}} \end{cases}$$

9.(c) 正则动量:

$$p_\theta = \frac{\partial L}{\partial \dot{\theta}} = m R^2 \dot{\theta}$$

体系的 Hamilton 量:

$$H = \dot{\theta} p_\theta - L = -\frac{1}{2} m \Omega^2 R^2 \sin^2 \theta + \frac{1}{2} m R^2 \dot{\theta}^2 + mgR \cos \theta$$

9.(d) 由于 Hamilton 量不显含时间：

$$\begin{aligned}\frac{dH}{dt} &= \frac{\partial H}{\partial t} + \frac{\partial H}{\partial \theta} \dot{\theta} + \frac{\partial H}{\partial p_\theta} \dot{p}_\theta \\ &= \frac{\partial H}{\partial t} + \frac{\partial H}{\partial \theta} \frac{\partial H}{\partial p_\theta} + \frac{\partial H}{\partial p_\theta} \left(-\frac{\partial H}{\partial \theta} \right) \\ &= \frac{\partial H}{\partial t} \\ &= 0\end{aligned}$$

Hamilton 量的数值是守恒量。

由于体系的 Hamilton 量：

$$H = -\frac{1}{2}m\Omega^2 R^2 \sin^2 \theta + \frac{1}{2}mR^2 \dot{\theta}^2 + mgR \cos \theta$$

而体系的总能量：

$$E = T + V = \frac{1}{2}m\Omega^2 R^2 \sin^2 \theta + \frac{1}{2}mR^2 \dot{\theta}^2 + mgR \cos \theta$$

故 $H \neq T + V$ 。